

Relational Database Management Systems and Schema Normalization

Student Name: Priya Patel | Roll Number: CS-2024-019

1. Relational Database Concepts

A relational database management system organizes enterprise data into tables consisting of rows and columns with primary keys and foreign keys enforcing referential integrity across schemas.

2. Transaction Reliability and ACID Properties

ACID properties ensure database transaction reliability: Atomicity guarantees all-or-nothing execution, Consistency enforces semantic integrity, Isolation prevents concurrent collision, and Durability ensures committed updates persist permanently.

3. Schema Normalization

Database normalization systematically restructures relations to eliminate redundant storage and insertion, update, or deletion anomalies across First, Second, Third, and Boyce-Codd Normal Forms.